

8.4 Reparameterization and side conditions

A rank-deficient model can be handled in three ways. One can work with it as it stands, estimating only estimable functions, which is what the last two sections did. One can *reparameterize*, replacing $\mathbf{X}$ by a full-rank matrix with the same column space. Or one can impose *side conditions*, linear restrictions on $\boldsymbol{\beta}$ that select a single coefficient vector from each set $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$. This section shows that the second and third are two descriptions of the same act: choosing coordinates for the mean vector. Neither changes any estimable quantity. It also records what the most common software does by default.

8.4.1 Full-rank reparameterizations

By Corollary 8.2.3, at most r estimable functions are linearly independent. A set of exactly r of them can serve as a new parameter vector.

Theorem 8.4.1 — Reparameterizing to full rank

Let $\text{rank}(\mathbf{X}) = r$. Let $\mathbf{U}$ be an $r \times p$ matrix of rank r whose rows lie in $\mathcal{C}(\mathbf{X}^\top)$, so that $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$ is a vector of r linearly independent estimable functions, and put

$$\mathbf{Z} = \mathbf{X}\mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}. \quad (8.4.1)$$

Then:

- (a) $\mathbf{X} = \mathbf{Z}\mathbf{U}$, $\mathbf{Z}$ has full column rank r , and $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$;
- (b) the models $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $\boldsymbol{\beta} \in \mathbb{R}^p$, and $E(\mathbf{Y}) = \mathbf{Z}\boldsymbol{\gamma}$, $\boldsymbol{\gamma} \in \mathbb{R}^r$, describe the same set of mean vectors, with $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$;
- (c) the least squares estimate in the new model is $\hat{\boldsymbol{\gamma}} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{y} = \mathbf{U}\hat{\boldsymbol{\beta}}$ for every least squares solution $\hat{\boldsymbol{\beta}}$ of the old one;
- (d) $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ is estimable iff $\boldsymbol{\lambda} = \mathbf{U}^\top\mathbf{c}$ for some $\mathbf{c} \in \mathbb{R}^r$, which is then unique, and in that case $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\hat{\boldsymbol{\gamma}}$;
- (e) conversely, if $\mathbf{Z}$ is any $n \times r$ matrix of full column rank with $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$, then $\mathbf{X} = \mathbf{Z}\mathbf{U}$ for a unique $\mathbf{U}$, and $\mathbf{U}$ has rank r and estimable rows.

Proof. Since $\mathcal{C}(\mathbf{U}^\top) \subseteq \mathcal{C}(\mathbf{X}^\top)$ and both have dimension r , $\mathcal{C}(\mathbf{U}^\top) = \mathcal{C}(\mathbf{X}^\top)$. The matrix $\mathbf{Q} = \mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}\mathbf{U}$ is symmetric and idempotent with column space $\mathcal{C}(\mathbf{U}^\top)$, so it is the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$ (Theorem 6.3.3). Hence $\mathbf{I} - \mathbf{Q}$ projects onto $\mathcal{C}(\mathbf{X}^\top)^\perp = \mathcal{N}(\mathbf{X})$, and $\mathbf{X}(\mathbf{I} - \mathbf{Q}) = \mathbf{0}$, that is, $\mathbf{X} = \mathbf{X}\mathbf{Q} = \mathbf{Z}\mathbf{U}$. (a) Now $r = \text{rank}(\mathbf{X}) = \text{rank}(\mathbf{Z}\mathbf{U}) \leq \text{rank}(\mathbf{Z}) \leq r$, so $\mathbf{Z}$ has full column rank. $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X})$ because $\mathbf{Z} = \mathbf{X}(\cdot)$, and $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ because $\mathbf{X} = \mathbf{Z}\mathbf{U}$. (b) $\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}(\mathbf{U}\boldsymbol{\beta})$, and $\boldsymbol{\beta} \mapsto \mathbf{U}\boldsymbol{\beta}$ maps $\mathbb{R}^p$ onto $\mathbb{R}^r$ because $\mathbf{U}$ has full row rank. So both models describe $\mathcal{C}(\mathbf{X})$. (c) $\mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Z}(\mathbf{U}\hat{\boldsymbol{\beta}})$ by Theorem 6.4.1, and $\mathbf{Z}$ has full column rank. (d) $\boldsymbol{\lambda}$ is estimable iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{U}^\top)$. The vector $\mathbf{c}$ is unique because $\mathbf{U}^\top$ has full column rank, and $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\mathbf{U}\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\hat{\boldsymbol{\gamma}}$. (e) Each column of $\mathbf{X}$ lies in $\mathcal{C}(\mathbf{Z})$, so $\mathbf{X} = \mathbf{Z}\mathbf{U}$ with $\mathbf{U} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{X}$, unique because $\mathbf{Z}$ has full column rank. Then $\mathbf{U}\boldsymbol{\beta} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top(\mathbf{X}\boldsymbol{\beta})$ is a linear function of $\mathbf{X}\boldsymbol{\beta}$, so each row of $\mathbf{U}$ is estimable. Finally $r = \text{rank}(\mathbf{X}) \leq \text{rank}(\mathbf{U}) \leq r$.

So reparameterization is the same thing as choosing a basis for the estimable space. Theorem 6.7.1 of Chapter 6 already showed that fitted values and residuals do not care which basis is chosen. Part (d) adds that estimable functions do not care either: each is a fixed linear combination of the new parameters, and its estimate is the same combination of their estimates. Everything in the full-rank theory of Chapter 5 now applies to $\hat{\mathbf{y}}$, including Theorem 5.5.1 and the variance estimator of Theorem 5.6.1, with p replaced by r .

For the one-way layout with $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_g)^\top$, two choices of $\mathbf{U}$ stand out. The rows $(1, \mathbf{e}_k^\top)$, $k = 1, \dots, g$, give $\boldsymbol{\gamma} = (\mu + \alpha_1, \dots, \mu + \alpha_g)^\top$, the cell means, and $\mathbf{Z}$ is the indicator matrix. The rows $(1, \mathbf{e}_1^\top)$ and $(0, \mathbf{e}_k^\top - \mathbf{e}_1^\top)$, $k = 2, \dots, g$, give the first group's mean and the differences from it. This is the reference coding of Section 8.5.

8.4.2 Side conditions

The other classical device leaves $\mathbf{X}$ alone and adds equations. A **side condition** is a set of linear restrictions $\mathbf{T}\mathbf{b} = \mathbf{0}$, with $\mathbf{T}$ a $q \times p$ matrix, imposed on the solutions of the normal equations. The aim is to single out one solution. Examples in the one-way layout are $\alpha_1 = 0$, $\sum_k \alpha_k = 0$ and $\mu = 0$. The theorem says exactly which restrictions achieve this.

Theorem 8.4.2 — Side conditions

Let $\mathbf{X}$ be $n \times p$ of rank r and $\mathbf{T}$ be $q \times p$. The following are equivalent:

- (a) for every $\mathbf{y} \in \mathbb{R}^n$ there is exactly one $\mathbf{b}$ with $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$;
- (b) for every $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ there is exactly one $\mathbf{b}$ with $\mathbf{X} \mathbf{b} = \boldsymbol{\mu}$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$;
- (c) $\mathbb{R}^p = \mathcal{N}(\mathbf{X}) \oplus \mathcal{N}(\mathbf{T})$;
- (d) $\text{rank} \begin{pmatrix} \mathbf{X} \\ \mathbf{T} \end{pmatrix} = p$ and $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$;
- (e) $\text{rank}(\mathbf{T}) = p - r$, and no nonzero linear combination of the rows of $\mathbf{T}$ is estimable.

When these hold, $\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T}$ is nonsingular, the solution in (a) is

$$\hat{\boldsymbol{\beta}}_{\mathbf{T}} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{y}, \quad (8.4.2)$$

and $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1}$ is a generalized inverse of $\mathbf{X}^\top \mathbf{X}$.

Proof. (a) $\Leftrightarrow$ (b). The solutions of the normal equations for $\mathbf{y}$ are the $\mathbf{b}$ with $\mathbf{X} \mathbf{b} = \mathbf{M} \mathbf{y}$ (Theorem 6.4.2), and $\mathbf{M} \mathbf{y}$ runs over all of $\mathcal{C}(\mathbf{X})$ as $\mathbf{y}$ does.

(b) $\Leftrightarrow$ (c). The set $\{\mathbf{b} : \mathbf{X} \mathbf{b} = \boldsymbol{\mu}\}$ is a coset $\mathbf{b}_0 + \mathcal{N}(\mathbf{X})$, and every $\mathbf{b}_0 \in \mathbb{R}^p$ arises (take $\boldsymbol{\mu} = \mathbf{X} \mathbf{b}_0$). So (b) says that every coset of $\mathcal{N}(\mathbf{X})$ meets $\mathcal{N}(\mathbf{T})$ in exactly one point. Some point $\mathbf{b}_0 + \mathbf{v}$ of the coset lies in $\mathcal{N}(\mathbf{T})$ iff $\mathbf{b}_0 \in \mathcal{N}(\mathbf{T}) + \mathcal{N}(\mathbf{X})$, so existence for every coset means $\mathbb{R}^p = \mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T})$. Two points of the same coset that both lie in $\mathcal{N}(\mathbf{T})$ differ by an element of $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T})$, and a nonzero element of that intersection added to one such point gives a second. So uniqueness means $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$. Together, these say the sum is direct.

(c) $\Leftrightarrow$ (d). By Lemma 6.2.1 and Lemma 6.2.5,

$$(\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}))^\perp = \mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top), \quad (\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}))^\perp = \mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top).$$

So $\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}) = \mathbb{R}^p$ iff $\mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top) = \{\mathbf{0}\}$, and $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$ iff $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top) = \mathbb{R}^p$. The last space is the row space of the stacked matrix, so this holds iff that matrix has rank p .

(d) $\Leftrightarrow$ (e). A vector in $\mathcal{C}(\mathbf{T}^\top)$ is a combination of the rows of $\mathbf{T}$, and it is estimable iff it lies in $\mathcal{C}(\mathbf{X}^\top)$. So the second halves of (d) and (e) say the same thing. When they hold, the sum $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$ is direct, with dimension $r + \text{rank}(\mathbf{T})$. This equals p iff $\text{rank}(\mathbf{T}) = p - r$.

For the formula, suppose (d). If $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{v} = \mathbf{0}$, then $0 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \mathbf{v}^\top \mathbf{T}^\top \mathbf{T} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 + \|\mathbf{T} \mathbf{v}\|^2$, so $\mathbf{v}$ lies in $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$. Hence the matrix is nonsingular. The solution $\mathbf{b}$ of (a) satisfies $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{b} = \mathbf{X}^\top \mathbf{y} + \mathbf{T}^\top \mathbf{0}$, which gives Eq. 8.4.2. Finally, write $\mathbf{G}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1}$. For every $\mathbf{y}$, $\mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{y}$ solves the normal equations, so $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top = \mathbf{X}^\top$. Multiplying on the right by $\mathbf{X}$ gives $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$.

■

Condition (e) is the familiar rule of thumb, made exact: a side condition must supply exactly as many restrictions as the rank deficiency, and none of them, singly or in combination, may be estimable. The reason for the second requirement is clear from (d). An estimable restriction constrains the mean vector itself, and so it changes the model instead of merely choosing coordinates.

Remark (Existence and uniqueness separately). The proof shows more than the theorem states. The condition $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$ alone guarantees that the restricted normal equations always have *at least* one solution. The condition $\text{rank}(\frac{\mathbf{X}}{\mathbf{T}}) = p$ alone guarantees *at most* one. A constraint that is too weak, such as a single restriction when $p - r = 2$, leaves many solutions. One with an estimable part leaves none for most $\mathbf{y}$ (Exercise (C2)).

8.4.3 What the constrained coefficients mean

Under a side condition every coefficient becomes identifiable, and it acquires a meaning.

Proposition 8.4.3 — Constrained coefficients are estimable functions

Suppose $\mathbf{T}$ satisfies the conditions of Theorem 8.4.2, and let $\mathbf{P}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{X}$. For each $\boldsymbol{\beta}$, $\mathbf{P}_\mathbf{T} \boldsymbol{\beta}$ is the unique vector $\mathbf{b}$ with $\mathbf{X} \mathbf{b} = \mathbf{X} \boldsymbol{\beta}$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$. The matrix $\mathbf{P}_\mathbf{T}$ is idempotent, with range $\mathcal{N}(\mathbf{T})$ and null space $\mathcal{N}(\mathbf{X})$. Every row of $\mathbf{P}_\mathbf{T}$ lies in $\mathcal{C}(\mathbf{X}^\top)$. So each constrained coefficient $\mathbf{e}_j^\top \mathbf{P}_\mathbf{T} \boldsymbol{\beta}$ is an estimable function of the original parameters, and $\hat{\boldsymbol{\beta}}_\mathbf{T}$ estimates it without bias.

Proof. With $\mathbf{y} = \mathbf{X} \boldsymbol{\beta}$, the vector $\mathbf{P}_\mathbf{T} \boldsymbol{\beta} = \mathbf{G}_\mathbf{T} \mathbf{X}^\top (\mathbf{X} \boldsymbol{\beta})$ is the unique solution of Theorem 8.4.2(a) for that $\mathbf{y}$. It satisfies $\mathbf{X} \mathbf{b} = \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{X} \boldsymbol{\beta}$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$. The matrix $\mathbf{P}_\mathbf{T}$ is $\mathbf{H}_\mathbf{G}$ of Eq. 8.3.1 for the generalized inverse $\mathbf{G}_\mathbf{T}$, so Proposition 8.3.3 gives idempotence,

the null space $\mathcal{N}(\mathbf{X})$ and rows in $\mathcal{C}(\mathbf{X}^\top)$. Its range lies in $\mathcal{N}(\mathbf{T})$. The range has dimension r , and so does $\mathcal{N}(\mathbf{T})$, because $\text{rank}(\mathbf{T}) = p - r$. So the two coincide. Unbiasedness is Proposition 8.3.3(c). ■

In words, $\mathbf{P}_\mathbf{T}$ is the projection of $\mathbb{R}^p$ onto $\mathcal{N}(\mathbf{T})$ along $\mathcal{N}(\mathbf{X})$, generally an oblique one. It sends the true $\boldsymbol{\beta}$ to the unique representative of $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$ that satisfies the side condition. That representative's coordinates are functions of the mean, so they can be estimated. Their *meaning* is fixed by $\mathbf{T}$. In the one-way layout, the condition $\sum_k \alpha_k = 0$ makes μ the unweighted average of the g cell means and α_k the deviation of cell k from it. The condition $\sum_k n_k \alpha_k = 0$ makes μ the size-weighted average instead. The condition $\alpha_1 = 0$ makes μ the first cell mean. The data do not choose among these. The analyst chooses, by deciding which question the coefficients should answer.

One side condition deserves special mention, because it is the one many programs impose without saying so.

Proposition 8.4.4 — Minimum norm as a side condition

Let the columns of the $p \times (p - r)$ matrix $\mathbf{V}_0$ be a basis of $\mathcal{N}(\mathbf{X})$. Then $\mathbf{T} = \mathbf{V}_0^\top$ satisfies the conditions of Theorem 8.4.2, and the resulting solution $\hat{\boldsymbol{\beta}}_\mathbf{T}$ is the minimum-norm least squares solution $\mathbf{X}^+\mathbf{y}$.

Proof. $\mathcal{C}(\mathbf{T}^\top) = \mathcal{C}(\mathbf{V}_0) = \mathcal{N}(\mathbf{X})$, which has dimension $p - r$ and meets $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ only in $\mathbf{0}$. So (d) holds. The least squares solutions are $\hat{\boldsymbol{\beta}}_\mathbf{T} + \mathcal{N}(\mathbf{X})$, and $\hat{\boldsymbol{\beta}}_\mathbf{T} \in \mathcal{N}(\mathbf{T}) = \mathcal{C}(\mathbf{V}_0)^\perp = \mathcal{N}(\mathbf{X})^\perp$. For $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, Pythagoras gives $\|\hat{\boldsymbol{\beta}}_\mathbf{T} + \mathbf{v}\|^2 = \|\hat{\boldsymbol{\beta}}_\mathbf{T}\|^2 + \|\mathbf{v}\|^2$, so $\hat{\boldsymbol{\beta}}_\mathbf{T}$ has the smallest norm. By Proposition 6.10.1 this is $\mathbf{X}^+\mathbf{y}$. ■

In the one-way layout the null space is spanned by $(1, -1, \dots, -1)^\top$, so the minimum-norm solution is the one satisfying $\mu = \alpha_1 + \dots + \alpha_g$. No one would choose this restriction on purpose, and its coefficients have no natural interpretation.

Example. (Five side conditions, one fit)

Eighteen simulated observations fall into four groups of sizes 5, 3, 6, 4, with group means 11.111, 15.204, 10.558 and 13.624. In the model $E(y_{kj}) = \mu + \alpha_k$, $p - r = 1$, so a side condition is a single nonestimable restriction. The listing solves Eq. 8.4.2 for five of them and computes the minimum-norm solution.

Side condition	$\hat{\mu}$	$\hat{\alpha}_1$	$\hat{\alpha}_2$	$\hat{\alpha}_3$	$\hat{\alpha}_4$
$\alpha_1 = 0$	11.111	0.000	4.093	-0.553	2.513
$\alpha_4 = 0$	13.624	-2.513	1.580	-3.066	0.000
$\sum_k \alpha_k = 0$	12.624	-1.513	2.580	-2.066	1.000
$\sum_k n_k \alpha_k = 0$	12.167	-1.056	3.037	-1.609	1.457
$\mu = 0$	0.000	11.111	15.204	10.558	13.624

minimum	10.100	1.012	5.105	0.459	3.525
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norm ($\mu = \sum_k \alpha_k$)

Every row solves the normal equations. In every row $\hat{\mu} + \hat{\alpha}_k$ is the k th group mean, and $\hat{\alpha}_1 - \hat{\alpha}_2$ is the same. The value of $\hat{\mu}$ is, in turn, the first group mean, the last group mean, the average of the four group means (12.624), the overall mean $\bar{y} = 12.167$, zero, and an uninterpretable 10.100.

Compare a restriction that *is* estimable, $\alpha_1 = \alpha_2$. It is not a side condition. Imposing it (by solving the Lagrange system for least squares under a linear restriction) forces the first two group means to be equal. The residual sum of squares rises from 14.665 to 46.078. The fit has changed, which is what testing the hypothesis $\alpha_1 = \alpha_2$ in [Chapter 11](#) measures.

```
import numpy as np

sizes = [5, 3, 6, 4]
groups = np.repeat(np.arange(4), sizes)
rng = np.random.default_rng(84)
y = np.array([12.0, 15.0, 11.0, 14.0])[groups] + rng.normal(scale=1.0, size=groups.size)
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # (mu, alpha_1..
                        alpha_4)
A, c = X.T @ X, X.T @ y
```

[Run this code online](#)

```
n_k = np.array(sizes, dtype=float)
side = {"alpha_1 = 0": [0, 1, 0, 0, 0],
        "alpha_4 = 0": [0, 0, 0, 0, 1],
        "sum alpha_k = 0": [0, 1, 1, 1, 1],
        "sum n_k alpha_k = 0": [0, *n_k],
        "mu = 0": [1, 0, 0, 0, 0]}
solutions = {}
for name, t in side.items():
    T = np.array([t], dtype=float)
    G = np.linalg.inv(A + T.T @ T) # (X'X + T'T)^{-1}: a g-inverse of X'X
    b = G @ c
    solutions[name] = b
    print(f"{name:20s} b = {np.round(b, 3)} T b = {(T @ b)[0]:.1e}")
b_mn = np.linalg.pinv(X) @ y
print(f"{'minimum norm':20s} b = {np.round(b_mn, 3)}",
      f" mu - sum(alpha) = {b_mn[0] - b_mn[1:].sum():.1e}")
```

[Run this code online](#)

```
# "alpha_1 = alpha_2" is a restriction on an estimable function: it changes the
model
T_bad = np.array([[0, 1, -1, 0, 0]], dtype=float)
K = np.block([[A, T_bad.T], [T_bad, np.zeros((1, 1))]]) # Lagrange (bordered)
system
b_bad = np.linalg.lstsq(K, np.append(c, 0.0), rcond=None)[0][:5]
sse_free = np.sum((y - X @ solutions["alpha_1 = 0"]) ** 2)
sse_bad = np.sum((y - X @ b_bad) ** 2)
```

```
print(f"SSE without restriction {sse_free:.3f}; with alpha_1 = alpha_2 {sse_bad:.3f}
      ")
```

[Run this code online](#)

Reparameterization and side conditions are two languages for the same choice. Deleting the column of α_1 from $\mathbf{X}$ and fitting the full-rank remainder gives exactly the solution with $\alpha_1 = 0$. More generally, deleting a set of columns whose removal leaves a full-rank matrix with the same column space is the side condition that sets those coefficients to zero, and it satisfies Theorem 8.4.2(c) (Exercise (B1)).

8.4.4 What software does

Programs treat a rank-deficient $\mathbf{X}$ differently.

- *Code the factor at full rank before fitting.* Formula interfaces such as R's `lm()` and statsmodels' formula API expand a factor with g levels into an intercept and $g - 1$ columns, by default the reference coding of Section 8.5. The overparameterized matrix never appears. This is a reparameterization in the sense of Theorem 8.4.1.
- *Drop columns found to be dependent.* If the columns supplied are dependent anyway, R's `lm()` computes a QR decomposition in which a column found to be (numerically) a combination of earlier ones is moved to the end and left out. Its coefficient is reported as NA. This is the side condition setting that coefficient to zero, and which column is dropped depends on the order of the columns. SAS's GLM procedure similarly solves the normal equations with a generalized inverse that zeroes the coefficients of dependent columns, and it flags the resulting estimates as not unique.
- *Return the minimum-norm solution.* `numpy.linalg.lstsq` returns $\mathbf{X}^+\mathbf{y}$ silently (it reports the numerical rank only as a second output), and statsmodels' OLS, whose `fit` uses the pseudoinverse by default, returns the same vector with a `SingularMatrixWarning`. This is the side condition of Proposition 8.4.4, and it does not depend on the order of the columns.
- *Pivot greedily.* A QR decomposition with column pivoting (Chapter 10, Theorem 10.7.2) chooses at each step the remaining column with the largest component orthogonal to those already chosen, and the columns left at the end are dropped. In Example (Five side conditions, one fit) it takes the intercept, then α_3 (the largest group), α_1 and α_4 , and drops α_2 , the smallest group, a choice that depends on the group sizes.

```
import scipy.linalg
import statsmodels.api as sm
b_sm = sm.OLS(y, X).fit().params          # statsmodels default: method="pinv"
"
b_np = np.linalg.lstsq(X, y, rcond=None)[0]  # LAPACK gelsd: minimum-norm
solution
_, R, piv = scipy.linalg.qr(X, mode="economic", pivoting=True)
print("statsmodels:", np.round(b_sm, 3))
print("numpy lstsq:", np.round(b_np, 3))
print("pivoted QR column order:", piv, " |R_kk| =", np.round(np.abs(np.diag(R)), 3))
```

[Run this code online](#)

The lesson of this section is that all these outputs are correct, and none is more correct than the others. They are different coordinate systems for one fitted mean vector. Any conclusion

that changes when the software changes is a conclusion about a nonestimable function.

Warning. Before interpreting a coefficient from a rank-deficient fit, find out which side condition produced it. A coefficient reported as “the effect of level 3” may be a difference from level 1, a deviation from an average of cell means, a deviation from the overall mean, or, under the minimum-norm convention, a quantity with no interpretation at all.

8.4.5 Exercises

8.4.5.1 A. Check your understanding

Exercise A1.

In the one-way model with four groups and $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_4)^\top$, decide which of the following are side conditions in the sense of [Theorem 8.4.2](#): (i) $\alpha_2 = 0$; (ii) $\mu = \alpha_3$; (iii) $\alpha_1 - \alpha_2 = 0$; (iv) $\alpha_1 = 0$ together with $\alpha_2 = 0$; (v) $2\mu + \alpha_1 + \alpha_2 = 0$.

8.4.5.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, where $\mathbf{X}_1$ has full column rank and $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$. Show that the least squares fit using $\mathbf{X}_1$ alone, padded with zeros for the coefficients of $\mathbf{X}_2$, is the solution of the side condition $\mathbf{T} = [\mathbf{0}, \mathbf{I}]$. Check the conditions of [Theorem 8.4.2](#) directly.

Solution. The padded vector $(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top$ satisfies $\mathbf{X}(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{M}\mathbf{y}$, because $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$. So it solves the normal equations and satisfies $\mathbf{T}\mathbf{b} = \mathbf{0}$. For the conditions: $\text{rank}(\mathbf{T}) = p_2$, and $\text{rank}(\mathbf{X}) = \text{rank}(\mathbf{X}_1) = p_1$, so $\text{rank}(\mathbf{T}) = p - r$. If $\mathbf{T}^\top \mathbf{c} = (\mathbf{0}^\top, \mathbf{c}^\top)^\top$ were in $\mathcal{C}(\mathbf{X}^\top)$, say $\mathbf{X}_1^\top \boldsymbol{\rho} = \mathbf{0}$ and $\mathbf{X}_2^\top \boldsymbol{\rho} = \mathbf{c}$, then $\boldsymbol{\rho} \perp \mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X}) \supseteq \mathcal{C}(\mathbf{X}_2)$, so $\mathbf{c} = \mathbf{0}$. Condition (e) holds.

Exercise B2.

For the additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$ with $i = 1, 2, j = 1, 2, 3$ and one observation per cell, take $\mathbf{U}$ with rows giving $\mu + \alpha_1 + \beta_1, \alpha_2 - \alpha_1, \beta_2 - \beta_1$ and $\beta_3 - \beta_1$. Compute $\mathbf{Z}$ from [Eq. 8.4.1](#), and check that it is the reference-coded model matrix.

Exercise B3.

Let $\mathbf{T}$ satisfy the conditions of [Theorem 8.4.2](#) and have full row rank $p - r$. Show that the bordered matrix $\begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{T}^\top \\ \mathbf{T} & \mathbf{0} \end{pmatrix}$ is nonsingular, that the solution of the bordered system with right-hand side $(\mathbf{X}^\top \mathbf{y}, \mathbf{0})$ has first block $\hat{\boldsymbol{\beta}}_{\mathbf{T}}$ and second block $\mathbf{0}$, and interpret the zero Lagrange multiplier. Relate this to Exercise (B2).

8.4.5.3 C. Going deeper

Exercise C1.

Replace $\mathbf{X}$ by $\mathbf{XD}$ for a nonsingular diagonal $\mathbf{D}$ (a change of units). Show that the minimum-norm solution for $\mathbf{XD}$, converted back to the original units, is the solution for the side condition $\mathbf{V}_0^\top \mathbf{D}^{-2} \mathbf{b} = \mathbf{0}$, which differs in general from $\mathbf{V}_0^\top \mathbf{b} = \mathbf{0}$. Conclude that the minimum-norm convention depends on the units of measurement, while no estimable function does.

Exercise C2.

Suppose $\text{rank}(\mathbf{T}) = p - r$ but $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) \neq \{\mathbf{0}\}$. Show that for some $\mathbf{y}$ the restricted normal equations of [Theorem 8.4.2\(a\)](#) have no solution, and that whenever they have one they have infinitely many.

Solution. By the dimension count, $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$ has dimension less than $r + (p - r) = p$. So $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) \neq \{\mathbf{0}\}$, and any solution can be moved along a nonzero vector of that intersection. For existence, the proof of [Theorem 8.4.2](#) shows that a solution exists for every $\mathbf{y}$ iff $\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}) = \mathbb{R}^p$, which fails because its orthogonal complement $\mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top)$ is nonzero. So some coset $\mathbf{b}_0 + \mathcal{N}(\mathbf{X})$ misses $\mathcal{N}(\mathbf{T})$, and the response $\mathbf{y} = \mathbf{Xb}_0$ has no solution.